WISDOM AND CHANCE — COMBAT MECHANIC UPDATE GUIDE

For the Web Server / Web Client AI Agent
Date: March 22, 2026

This document describes critical changes to how Guardian and Quick Strike interact during combat resolution. These changes must be applied to the web version's game engine to match the mobile client's behavior.

1. SUMMARY OF CHANGES

BEFORE (Old Behavior):

- Quick Strike damage was applied IMMEDIATELY to opponent HP during card deployment, completely bypassing Guardian
- Guardian ONLY blocked combat damage (power difference) for the LOSER
- The WINNER's Guardian did nothing

AFTER (New Behavior):

- Quick Strike damage is ACCUMULATED during deployment but NOT applied to HP until combat resolution
- Guardian blocks ALL incoming damage — Quick Strike, power difference, or both combined
- Guardian works for BOTH the winner AND the loser
- Guardian is a universal damage shield

2. DETAILED COMBAT RESOLUTION FLOW

The combat resolution now follows this exact order:

Step 1: DEPLOY CARDS
  - Cards are played face-down, then revealed
  - Quick Strike cards ACCUMULATE their QS damage value but do NOT apply it to opponent HP yet
  - Old code: opp.hp -= qsDamage <-- REMOVE THIS
  - New code: just track ps.quickStrikeDamage += qsDamage

Step 2: CALCULATE POWER
  - Sum all deployed card power for each player
  - p1Power = sum of player 1's deployed cards' currentPower
  - p2Power = sum of player 2's deployed cards' currentPower

Step 3: CALCULATE GUARDIAN
  - Sum all Guardian trait values for each player
  - p1Guard = sum of player 1's Guardian cards' traitValue (default 3)
  - p2Guard = sum of player 2's Guardian cards' traitValue (default 3)
  - Include any ability-granted shield bonuses

Step 4: HEALING (applied before damage)
  - Apply Restoration healing to each player
  - hp = min(initialHP, hp + healingAmount)

Step 5: DETERMINE WINNER
  - If p1Power > p2Power: p1 wins, powerDiffDmg = p1Power - p2Power
  - If p2Power > p1Power: p2 wins, powerDiffDmg = p2Power - p1Power
  - If equal: draw, powerDiffDmg = 0

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Step 6: CALCULATE TOTAL INCOMING DAMAGE (NEW)
  For each player, calculate ALL incoming damage:

  p1TotalIncoming = p2QS + (winner === 'p2' ? powerDiffDmg : 0)
  p2TotalIncoming = p1QS + (winner === 'p1' ? powerDiffDmg : 0)

  This means:
  - The LOSER receives: power difference + opponent's QS
  - The WINNER receives: opponent's QS only (if any)
  - In a DRAW: each player only receives opponent's QS (if any)

Step 7: APPLY GUARDIAN AS UNIVERSAL SHIELD (NEW)
  Guardian absorbs from the total incoming damage:

  p1Blocked = min(p1Guard, p1TotalIncoming)
  p2Blocked = min(p2Guard, p2TotalIncoming)

  p1NetDmg = max(0, p1TotalIncoming - p1Guard)
  p2NetDmg = max(0, p2TotalIncoming - p2Guard)

  IMPORTANT: p1Blocked/p2Blocked is the ACTUAL amount blocked (capped
  at incoming damage), NOT the raw Guardian value. This prevents showing
  "blocked 5" when only 3 damage was incoming.

Step 8: APPLY NET DAMAGE TO HP
  p1.hp -= p1NetDmg
  p2.hp -= p2NetDmg


========================================================================
3. PSEUDOCODE (Complete Combat Resolution)
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function resolveCombat(p1, p2):
    // Accumulate values (QS already tracked during deployment)
    p1QS = p1.quickStrikeDamage
    p2QS = p2.quickStrikeDamage

    p1Power = sum(p1.deployed.map(c => c.currentPower))
    p2Power = sum(p2.deployed.map(c => c.currentPower))

    p1Guard = sum(p1.deployed.filter(c => c.trait == 'Guardian')
                              .map(c => c.traitValue || 3))
    p2Guard = sum(p2.deployed.filter(c => c.trait == 'Guardian')
                              .map(c => c.traitValue || 3))

    // Healing (before combat damage)
    if p1.healingAmount > 0:
        p1.hp = min(MAX_HP, p1.hp + p1.healingAmount)
    if p2.healingAmount > 0:
        p2.hp = min(MAX_HP, p2.hp + p2.healingAmount)

    // Determine winner
    winner = 'draw'
    powerDiffDmg = 0
    if p1Power > p2Power:
        winner = 'p1'
        powerDiffDmg = p1Power - p2Power
    else if p2Power > p1Power:
        winner = 'p2'
        powerDiffDmg = p2Power - p1Power
```

```
    // Calculate total incoming damage for each player
    p1TotalIncoming = p2QS + (powerDiffDmg if winner == 'p2' else 0)
    p2TotalIncoming = p1QS + (powerDiffDmg if winner == 'p1' else 0)

    // Guardian blocks ALL incoming damage
    p1Blocked = min(p1Guard, p1TotalIncoming)
    p2Blocked = min(p2Guard, p2TotalIncoming)

    p1NetDmg = max(0, p1TotalIncoming - p1Guard)
    p2NetDmg = max(0, p2TotalIncoming - p2Guard)

    // Apply net damage
    p1.hp -= p1NetDmg
    p2.hp -= p2NetDmg
    p1.hp = max(0, p1.hp)
    p2.hp = max(0, p2.hp)

    // The "damage" field in RoundResult for backward compatibility
    damage = p2NetDmg if winner == 'p1' else (p1NetDmg if winner == 'p2' else 0)

    return RoundResult {
        p1Power, p2Power, winner, damage,
        p1HP: p1.hp, p2HP: p2.hp,
        p1QS, p2QS,
        p1Guardian: p1Blocked,  // Amount ACTUALLY blocked (not raw)
        p2Guardian: p2Blocked,
        p1Healing, p2Healing,
        p1NetDmg,  // NEW: actual HP lost by p1
        p2NetDmg,  // NEW: actual HP lost by p2
    }
```

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4. WORKED EXAMPLES
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EXAMPLE 1: Winner has Guardian, Loser has QS
--------------------------------------------
P1 deploys: Card A (power 10), Card B (power 8, Guardian 3)
P2 deploys: Card C (power 6, Quick Strike 4), Card D (power 5)


p1QS = 0, p2QS = 4
p1Guard = 3, p2Guard = 0

p1TotalIncoming = 4 (p2's QS) + 0 (p1 won) = 4
p2TotalIncoming = 0 (p1's QS) + 7 (power diff) = 7

p1Blocked = min(3, 4) = 3
p2Blocked = min(0, 7) = 0




Result: P1 wins. P2 takes 7 dmg. P1 takes 1 dmg (from QS, 3 blocked).


EXAMPLE 2: Loser has Guardian blocking QS + combat
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P1 deploys: Card A (power 12, Quick Strike 3)
P2 deploys: Card B (power 14), Card C (power 2, Guardian 5)
```

```
p1QS = 3, p2QS = 0
p1Guard = 0, p2Guard = 5

p1TotalIncoming = 0 (p2's QS) + 4 (power diff) = 4
p2TotalIncoming = 3 (p1's QS) + 0 (p2 won) = 3

p1Blocked = min(0, 4) = 0


p1NetDmg = max(0, 4 - 0) = 4


Result: P2 wins. P1 takes 4 dmg. P2 takes 0 (Guardian absorbed all QS).


EXAMPLE 3: Draw with QS and Guardian
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P1 deploys: Card A (power 8, Quick Strike 3), Card B (power 4, Guardian 2)
P2 deploys: Card C (power 7, Quick Strike 5), Card D (power 5)


p1QS = 3, p2QS = 5
p1Guard = 2, p2Guard = 0

p1TotalIncoming = 5 (p2's QS) + 0 (draw) = 5
p2TotalIncoming = 3 (p1's QS) + 0 (draw) = 3

p1Blocked = min(2, 5) = 2
p2Blocked = min(0, 3) = 0

p1NetDmg = max(0, 5 - 2) = 3
p2NetDmg = max(0, 3 - 0) = 3

Result: Draw. P1 takes 3 dmg (5 QS - 2 blocked). P2 takes 3 dmg.


EXAMPLE 4: Guardian fully blocks everything
--------------------------------------------
P1 deploys: Card A (power 6, Guardian 5), Card B (power 4, Guardian 3)
P2 deploys: Card C (power 8, Quick Strike 2)


p1QS = 0, p2QS = 2
p1Guard = 8, p2Guard = 0

p1TotalIncoming = 2 (p2's QS) + 0 (p1 won) = 2
p2TotalIncoming = 0 (p1's QS) + 2 (power diff) = 2


p2Blocked = min(0, 2) = 0


p2NetDmg = max(0, 2 - 0) = 2

Result: P1 wins. P2 takes 2 dmg. P1 takes 0 (Guardian absorbed all).
Guardian displayed as "2 blocked" (not 8 — capped at incoming damage).


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5. RoundResult INTERFACE CHANGES
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The RoundResult interface now includes two new fields:

```
interface RoundResult {
    round: number;
    p1Power: number;
    p2Power: number;
    winner: 'p1' | 'p2' | 'draw';
    damage: number;         // Backward compat: loser's net damage
    p1HP: number;
    p2HP: number;
    p1Cards: DeployedCard[];
    p2Cards: DeployedCard[];
    p1QS: number;           // Total QS damage p1 dealt
    p2QS: number;           // Total QS damage p2 dealt
    p1Guardian: number;     // Amount p1's Guardian ACTUALLY blocked
    p2Guardian: number;     // Amount p2's Guardian ACTUALLY blocked
    p1Healing: number;
    p2Healing: number;
    p1NetDmg: number;       // NEW: actual HP lost by p1 after Guardian
    p2NetDmg: number;       // NEW: actual HP lost by p2 after Guardian
    abilityEffects?: AbilityEffect[];
}
```

IMPORTANT: p1Guardian/p2Guardian now represents the amount ACTUALLY
blocked (capped at total incoming damage), NOT the raw Guardian value.
This is correct for display purposes.

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6. UI BREAKDOWN FORMAT
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The UI should display damage breakdowns as follows:

For the loser (non-draw):
  Main: "{loserNetDmg}dmg"
  Breakdown: "{powerDiff} power +{winnerQS} QS -{loserGuardianBlocked} blk"

  Only show breakdown if QS or Guardian was involved.
  Only show each part if its value > 0.

For the winner taking QS back-damage:
  Additional line: "+{winnerNetDmg} QS back"
  (Only shown if winner actually took net damage from opponent's QS)

For draws with QS:
  Show "QS resolves" with per-player damage

Combat log text examples:
  "Player wins! Opponent takes 5 dmg (7 power +3 QS -5 blocked)"
  "Player takes 1 dmg from opponent QS (2 blocked)"
  "Draw! Quick Strike damage still resolves."

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7. KEY RULES TO REMEMBER
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1. Guardian blocks EVERYTHING — QS, combat damage, or both
2. Guardian works for BOTH winner and loser

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3. QS is NOT applied immediately — it's resolved at combat time
4. Guardian blocked amount is CAPPED at incoming damage (display only
   the amount actually absorbed)
5. Both players can take damage in the same round (loser from combat+QS,
   winner from opponent's QS minus their own Guardian)
6. In a draw, only QS damage applies (no power difference), and each
   player's Guardian absorbs their incoming QS damage
7. The "damage" field in RoundResult is kept for backward compatibility
   and equals the LOSER's net damage (for non-draw rounds)
```